Werner **Bisschoff**

CLOUD INFRASTRUCTURE ENGINEER / PLATFORM DEVELOPER

AWS Certified Solutions Architect – Associate (In Progress — Target: 6 Weeks)

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Cloud Infrastructure Engineer / Platform Developer with 5+ years of experience designing high-availability infrastructure across Linux systems administration, Docker containerization, and Pulumi infrastructure-as-code. Proven track record in PostgreSQL performance tuning, multi-tenant data isolation, and building resilient automation and deployment pipelines.

Skills

Primary Competencies : Python, Docker, Docker Compose, AWS (Solutions Architect – In Progress), PostgreSQL (Row-Level Security, Index Tuning, Connection Pooling)
Secondary Platforms : ERPNext/Frappe Framework, Elixir/OTP, Asynchronous Workflow Engines (Inngest, Oban)
Data Layer Infrastructure : SQL, Ecto.Multi
Systems & Automation : Pulumi (IaC), Linux Systems Administration, CI/CD Pipelines, Bash Scripting, Network Proxies, SSH Networking, Spec-Driven Development (SDD)

Experience

Principal Systems Architect — Cape Town

DIVERGENT TABLETOP — Jul 2025

- Engineered and hosted an enterprise-grade documentation ecosystem.
- Implemented a single-database multi-tenant separation engine using PostgreSQL Row-Level Security (RLS) and transaction-scoped local parameters (SET LOCAL), enforcing strict data silos across isolated groups.
- Designed an atomic asynchronous flush strategy using in-memory GenServer storage buffers, reducing database persistent write transactions by 85% during periods of high user interaction.
- Utilized Ecto.Multi atomic chains and Oban transactional background workers to ensure multi-stage backend edits (e.g., structural document renaming and incoming backlink indexing) execute completely or roll back safely.
- Applied Universal Design and Executive Functioning support models to product UX/UI; established highly structured, low-context asynchronous operational playbooks that minimized cognitive friction for contributors.

Full-Stack ERPNext & Automation Engineer — Cape Town

FARO AFRICA — Aug 2024 – Nov 2025

- Identified poor UX as root cause of recurring user errors — pushed back against 'training gap' framing, proposed user research to validate, and drove UX improvements that reduced task completion time and error rates.

  Internal warehouse tools were hard to use — requiring many taps to navigate to common actions. There had been frequent user error reports. The tech team attributed this to a "training gap," but I identified it as a UX problem. Changes were initially resisted as "unnecessary."

  Improve the UX of internal tools during a Retool-to-Expo migration.

  I pushed back on the "training gap" framing, arguing the root cause was poor UX design. I proposed interviewing ground staff to validate my hypothesis, then made initial improvements based on gut calls to demonstrate value. I took ownership of the UX design for the new versions and argued for the investment.

  The team eventually adopted the UX changes. Fewer taps reduced task completion time and user error rates dropped — validating that the problem was UX, not training.

- Refactored and extended ERPNext via Python and JavaScript server hooks, optimizing pricing matrices and BI reporting.
- Architected migration of internal Retool workflows to Expo/React Native, reducing vendor licensing overhead.
- Built custom NFC scanner utilities within Expo for instantaneous physical inventory syncs.
- Provisioned and maintained a secure virtualized Linux infrastructure environment using Docker Compose for container isolation and writing custom backup and monitoring scripts.
- Diagnosed and resolved issues in a large existing ERPNext installation; introduced LLM-assisted debugging workflows.

Enterprise Infrastructure & Tooling Engineer
Remote

Ingenics Digital GmbH (through ViVa Outsourcing)
Mar 2023 – May 2024

- Built modular Python utility structures and automation scripts for internal configuration schema validation.
- Designed data handling pipelines ensuring data transfer consistency across runtime platforms using declarative specifications.
- Developed Python hardware mocks enabling faster development cycles with fewer hardware dependencies.

Infrastructure & Tooling Developer
Cape Town

UMAN Technologies
Mar 2021 – Dec 2022

- Proactively created a Docker development container replicating the exact hardware environment — including VLAN and networking configuration — reducing setup time from hours to under an hour and enabling remote teams to develop without physical hardware.

  Remote teams were developing software for an embedded Linux hardware system. Each developer set up their own environment manually, leading to inconsistencies and slow onboarding.

  Enable the team to develop and test reliably without physical hardware, regardless of local setup.

  I proactively created a Dockerfile that replicated the exact hardware environment — including VLAN setup and networking configuration — matching production precisely. I had to convince a skeptical team member who preferred pushing images to a remote service, arguing for a simpler local-first approach.

  Setup time reduced from hours to under an hour. Remote teams could run and test the full system without physical hardware. Onboarding new engineers became significantly faster.

- Built Python hardware mocks simulating SOME/IP behavior; integrated into CI to enable automated testing without physical hardware, improving test coverage and developer productivity.

  Physical hardware wasn't always available for testing. Automated tests couldn't run at all.

  Enable automated testing without access to real hardware.

  I built Python mocks that simulated SOME/IP requests and responses, creating a test harness that replicated hardware behavior. I ensured the mocks were integrated into CI so tests ran automatically against them.

  The mocks were adopted by the team and integrated into CI. Automated tests could run without physical hardware, improving test coverage and developer productivity.

Education

B.Eng. Computer and Electronic Engineering

North-West University
2020

Projects

Divergent Tabletop Wiki

- Built a community wiki using Astro, Elixir, and Docker for knowledge management. Documented event frameworks, onboarding processes, and communication best practices. Created tooling for content management and community operations.

Ingenics Digital GmbH — Event-Driven FSM for Embedded Systems

- Designed event-driven finite state machine for I2C-based embedded system using C++ and FreeRTOS. Solved complex state management challenges in real-time embedded environment. Outcome: Maintainable in-house architecture leading to fewer bugs and quicker development cycles.